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Interpretable Lung Cancer Detection using Explainable AI Methods

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Abstract—Lung cancer is the second most frequently diagnosed cancer worldwide. It is one of the major causes of death among both men and women. In order to reduce the death rate, early detection of lung cancer is the only approach. In this paper, we have proposed an interpretable lung cancer diagnostic system using different ML models like Decision Tree, Logistic Regression, Random Forest, and Naive Bayes classifier. A Kaggle dataset named ‘Lung Cancer Detection’ is used to perform our research work. Among different ML models, Logistic regression and Random forest classifiers scored the highest accuracy of 97% in predicting lung cancer. We have also used the two most popular XAI models, SHAP and LIME to show the interpretability of our used models.

Index Terms—Decision Tree, Logistic Regression, Explainable AI, Random Forest, SHAP, LIME

I. INTRODUCTION

The second most widespread cancer worldwide is lung cancer. The only approach to treat lung cancer is through early detection as the reason of lung cancer is unclear and it's prevention is quite impossible. According to the WCRF [1], there are more than 2.2 million lung cancer cases in the world in 2020. Another article [2] estimated that in the U.S, 236,740 persons will be diagnosed with lung cancer by 2022. Smoking is the leading cause of lung cancer. However, this is not the best feature for detecting lung cancer. The only way to prevent lung cancer and reduce death due to lung cancer is early detection of it.

Many researchers have already used different ML models to predict lung cancer. Machine learning algorithms help to predict and classify different types of images and signals. Deep learning methods have the drawback that the parameters must be manually created in order to evaluate the best performance which makes it challenging to reproduce better outcomes [3]. A very few studies have focused on the interpretability of different models' decisions while detecting lung cancer and most of those works are image based. It is critical to identify the features that are responsible for predicting lung cancer since lives will be at stake.

In our research paper, we used various Machine Learning models to identify lung cancer based on patient symptoms. In this study, we employ Decision Tree, Logistic Regression,

Random Forest, and Naive Bayes classifiers as the ML models. However, the main focus of our study is not only the precise diagnosis of lung cancer, but also the explainability of the diagnosis. To accomplish this objective, we used SHAP and LIME to explain the predicted output of our ML models.

The following are the paper's significant contributions: We introduce a system capable of accurately diagnosing lung cancer with Machine Learning models and giving model interpretability with XAI models. For diagnosing lung cancer, we employ four distinct Machine Learning models. By using two XAI models termed SHAP and LIME, we demonstrate the explainability of our top two ML models. We identified two sets of features as the most relevant for lung cancer detection and interpretation.

II. RELATED WORK

A research paper by Kumar et al. (2022) [4] proposed a model to predict lung cancer from an image dataset. In their research, they used an ML model based on Support Vector Machine to classify the patients depending on their symptoms. A dataset, collected from an online repository of the University of California is used to train and test the model. At first, they performed some preprocessing tasks in the dataset like deleting unnecessary values. After splitting the dataset into train: test, they extracted features from the dataset and used the SVM model to predict the development of lung cancer. Finally, they compared their model with other existing models like KNN, Naive Bayes, and J48 and achieved a better result with an accuracy of around 98.8%.

Chehade et al. (2022) [5] implemented a CAD system to analyze colon cancer and lung cancer using histopathological images collected from the LC25000 dataset. GLCM method and Hu invariant moments are used to extract features from the images. They have applied MLP, XGBoost, RF, SVM, and LDA models to classify colon and lung cancer. Moreover, the SHAP method is used to explain the predicted outputs of the ML models. Among the five models, XGBoost achieved the

highest F1 score and accuracy of 98.8% and 99% respectively.

Another article [6] proposed a model to detect lung cancer in the human body based on its symptoms. They have used the ANN model in their research work to predict lung cancer. A dataset collected from Kaggle named ‘survey lung cancer’ is used to train their model. The dataset contains different symptoms like anxiety, coughing, fatigue, allergy, etc for both men and women of different age ranges. Their applied model predicts lung cancer accurately with an accuracy of 96.67%.

In a research article [7], the author proposed a method to find out the early stage of lung cancer using different Machine learning models. For their research, they have used three image datasets TCIA Dataset, LIDC-IDRI, and Kaggle dataset. At first, they used the U-Net CNN model for lung nodule segmentation. After that, they extracted features and used the CNN model to classify cancer and non-cancer. They compared their model with other available models and achieved an AUC of 0.6459.

Li et al. (2020) [8] suggested an interpretable lung cancer prediction system where they used CNN models. Their applied method consists of two modules: Deep classification network and estimation network. They have used the LUNA16 dataset to predict lung cancer. They compared their proposed method’s performance with some other baseline models like VBP, CAM, and LRP. Their proposed method’s classification accuracy on ResNet-18 is 82.15%.

III. METHODOLOGY

A. Dataset Description and Preprocessing

For our research work, we have collected a dataset named ‘Lung Cancer Detection’ from Kaggle. The dataset contains different cancer symptoms of 309 patients of different age ranges. Among the 309 patients, 162 are Male (M) and 147 are Female (F). The age of the Female patients is between 21 to 87 years and for the Male, the age range is from 39 to 79 years. The symptoms used in the dataset are smoking, yellow fingers, Alcohol, Fatigue, Shortness of Breath, Coughing, Chest Pain, etc. In the dataset, all the symptoms are classified into two labels: YES (2) and NO (1). Based on the symptoms, it predicts if the patient has lung cancer then it will show YES, otherwise NO.

Before applying different models for lung cancer prediction, we performed some preprocessing steps on our dataset. Though our dataset contains symptoms of 309 patients, among them 33 values are duplicates. So in the data preprocessing step, we first removed duplicate values. After removing duplicate values, our dataset contains 276 unique values. All the symptoms are classified into two labels YES and NO. So for our convenience, we have encoded the categorical data into numeric values 0 (NO) and 1 (YES). There is an imbalance between the cancerous (238) and non-cancerous (28) values in

our dataset. So we applied smote in the preprocessing step for oversampling of our dataset. After applying smote, our dataset contains 238 samples for both cancerous and non-cancerous patients.

B. Description Classification Models

1) *Decision Tree*: Decision tree is the most effective and well known method for categorization and prediction problems. It is organized hierarchically and has a root node, branches, internal nodes, and leaf nodes. Its structure is similar to a tree, and it is used to build a model that predicts the target value based on some decisions obtained from the training dataset [9]. A decision tree can handle both numeric and categorical data but it is very popular due to its good performance on categorical features. As our dataset contains categorical values, we have applied Decision Tree classifier in our research.

2) *Logistic Regression*: Logistic regression is a supervised machine learning model used in classification tasks. It is based on the concept of probability. Based on a few dependent variables, it estimates the probabilities of a few classes. The goal of this classifier is to predict the relationship between one categorical dependent variable with one or more independent variables. A tree based model always works best with categorical data [10]. Therefore, since logistic regression is based on the concept of tree-based models, we applied it to our dataset.

3) *Random Forest*: A machine learning technique called Random Forest (RF) uses the idea of Decision Trees to classify data. It can perform both classification and regression tasks. The quantity of decision trees determines how accurate a prediction is. If the number of decision trees increases, then the accuracy of prediction will be higher [11]. One of the most important aspects of Random forest is that it can handle categorical data for classification tasks with more accurate results. This is one of the reasons we choose Random Forest as one of our ML models.

4) *Naive Bayes*: Naive Bayes classifier is another supervised machine learning model used in classification tasks. It is based on the Bayes theorem and it is referred to as naïve because it disregards the prior distribution of parameters and assumes that all features are independent [12]. In comparison to other classification methods, Naive Bayes is a lot faster and works better with categorical data than numeric variables. Due to having categorical features in our dataset, we have applied Naive Bayes for classification purposes.

C. Description Interpretable Models

1) *SHAP*: In our research, we used SHAP (SHapley Additive exPlanations) values to demonstrate the interpretability and transparency of the machine learning

models we applied. SHAP offers global model interpretability [13]. It basically measures how much each feature contributes to the prediction. By calculating the Shapely values for the entire dataset, it can determine the global interpretation.

2) *LIME*: LIME is another interpretable model which is used in demonstrating the interpretability of machine learning models. Unlike SHAP, it offers local model interpretability. Instead of offering a global model interpretation, LIME suggests concentrating on local interpretation [14]. It changes one sample of data by updating the feature values, then tracks how the change affects the output. In our research, we have used LIME to correctly interpret the output of our used machine learning models.

IV. RESULTS AND DISCUSSION

Two of the four models we selected for the classification test performed admirably. The performance of the other two models was also satisfactory.

Table 1 shows that the accuracy for both the non-cancerous and malignant classes is 95% when using the decision tree model. The precision, recall, and F1 score for the Non-Cancerous class are 96%, 94%, and 95%, respectively, while they are 93%, 95%, and 94% for the malignant class. Our least accurate model, Naive Bayes, has a 92% accuracy across both classes. When employing Naive Bayes, the precision, recall, and F1 score for the Non-Cancerous class are 96%, 88%, and 92% respectively, while for the Cancerous class, the values are 88%, 95%, and 91%. With an accuracy of 97%, Logistic Regression and Random Forest are our top models. For both Logistic Regression and Random Forest models, the precision, recall, and F1 score for the non-malignant class are 96%, 98%, and 97%, respectively, whereas the values are 98%, 95%, and 97% for the malignant class.

To demonstrate explainability utilizing two XAI models, SHAP and LIME, we will use our top two machine learning (ML) models. A simple, theoretically valid way for explaining predictions for any model is created by SHAP by combining various existing approaches. The size and direction of a feature's impact on a prediction, whether positive or negative, are quantified by SHAP values. SHAP can also identify the overall interpretation by computing and combining the Shapely values for the full dataset. Therefore, we will employ SHAP to demonstrate the global interpretability of the selected ML models.

A global overview of the SHAP values dispersed throughout the Logistic Regression model's features is shown in Figure 1. The distribution of feature significance is also shown for each feature. The features are displayed in descending order of overall feature relevance, with the first feature being the most important and the last being the least. The significance or contribution of the features to the entire dataset may be better understood by referring to Fig. 1. For instance,

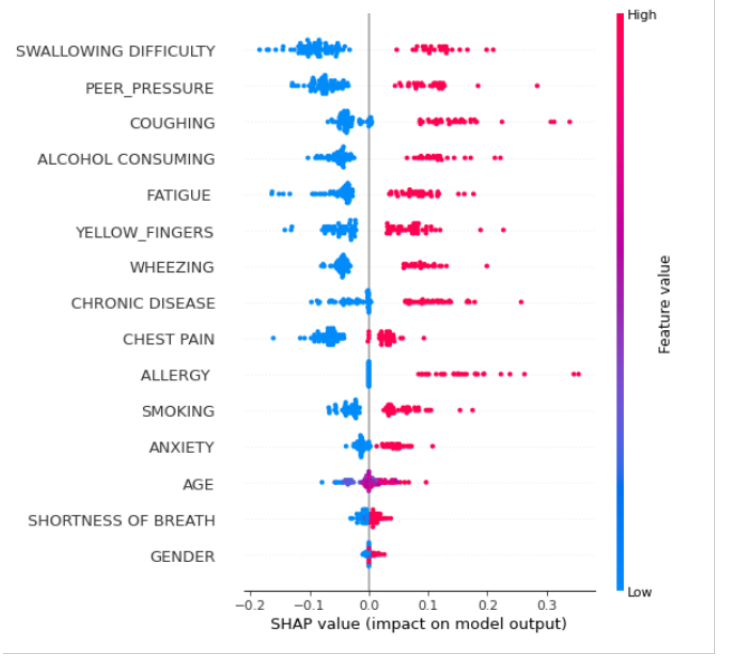


Fig. 1. Summary of SHAP values across all features for Logistic Regression

the Swallowing Difficulty feature's high values provide a significant positive contribution to the forecast, whereas its low values make a significant negative contribution. When its values are high, the Coughing variable likewise contributes greatly in the positive direction and when its values are low it makes a high negative contribution to the prediction. Regardless of how high or low their values are, the features Age and Gender have practically little impact on the prediction.

Fig. 2 displays a global view of the SHAP values distributed over the Random Forest model's features. With the first feature being the most significant and the final being the least, the features are presented in descending order of overall feature significance. The figure also shows the importance or contribution of the features to the entire dataset. For instance, the Allergy feature's high values provide a significant positive contribution to the forecast, whereas its low values make a significant negative contribution. The Swallowing Difficulty variable provides a significant positive contribution to the prediction when its values are high and a significant negative impact when its values are low. The features Age and Gender have practically no influence on the prediction, no matter how high or low their values are.

In contrast to SHAP, LIME offers local model interpretability. LIME modifies one sample of data by altering the feature values, then monitors the impact on the output. This frequently relates to the inquiries that individuals have while examining a model's output. We'll discuss LIME on our top ML models in the next part.

TABLE I
ACCURACY, PRECISION, RECALL AND F1-SCORE OF NON-CANCEROUS AND CANCEROUS CLASS FOR THE 4 DIFFERENT ML MODELS

Model Name	Non-Cancerous				Cancerous			
	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
Decision Tree	0.95	0.96	0.94	0.95	0.96	0.93	0.95	0.94
Logistic Regression	0.97	0.96	0.98	0.97	0.97	0.98	0.95	0.97
Random Forest	0.97	0.96	0.98	0.97	0.97	0.98	0.95	0.97
Naive Bayes	0.92	0.96	0.88	0.92	0.92	0.88	0.95	0.91

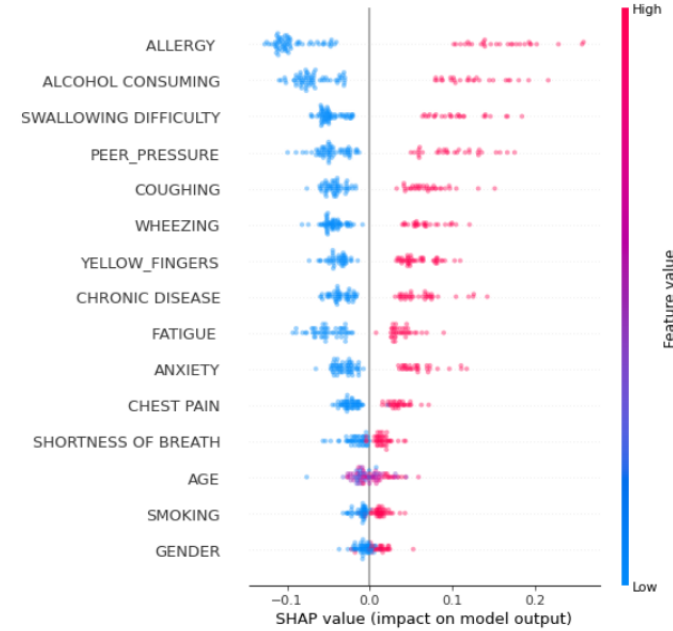


Fig. 2. Summary of SHAP values across all features for Random Forest

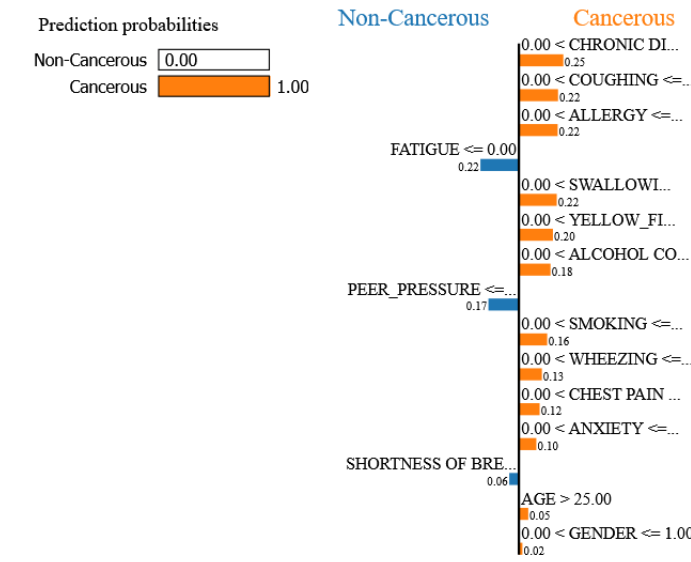


Fig. 3. The Logistic Regression model's LIME explainability for a single instance (Cancerous)

As seen in Figure 3, the Logistic Regression model predicts this specific instance as being Cancerous or class 1 with 100% confidence and provides an explanation for the prediction based on the features Chronic Disease, Coughing, Allergy, Swallowing Difficulty, Yellow Fingers, Alcohol Consuming, Smoking, Wheezing, Chest Pain, Anxiety, Age, and Gender. These features all go toward class 1, which is the Cancerous class. Out of all of these features, Chronic Disease, Coughing, Allergy, Swallowing Difficulty, and Yellow Fingers had the greatest influence on the prediction of the Cancerous class.

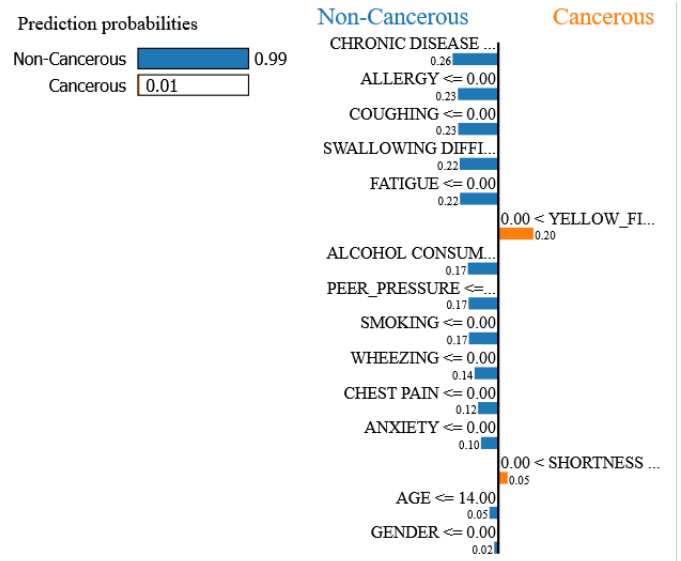


Fig. 4. The Logistic Regression model's LIME explainability for a single instance (Non-Cancerous)

As can be seen in Figure 4, the Logistic Regression model predicts that this particular occurrence will be non-cancerous or class 0 with 99% confidence and offers an explanation for the prediction based on all the parameters other than Yellow Finger and Shortness of Breath. All of these features converge toward class 0, or the non-cancerous class. Chronic Disease, Allergy, Coughing, Swallowing Difficulty, and Fatigue had the biggest impact on the prediction of the non-cancerous class out of all of these features.

Figure 5 illustrates how the Random Forest model, using the features Allergy, Alcohol Consuming, Swallowing Difficulty, Peer Pressure, Chronic Disease, Coughing, Yellow Fingers, Wheezing, Anxiety, and Shortness of Breath, predicts this

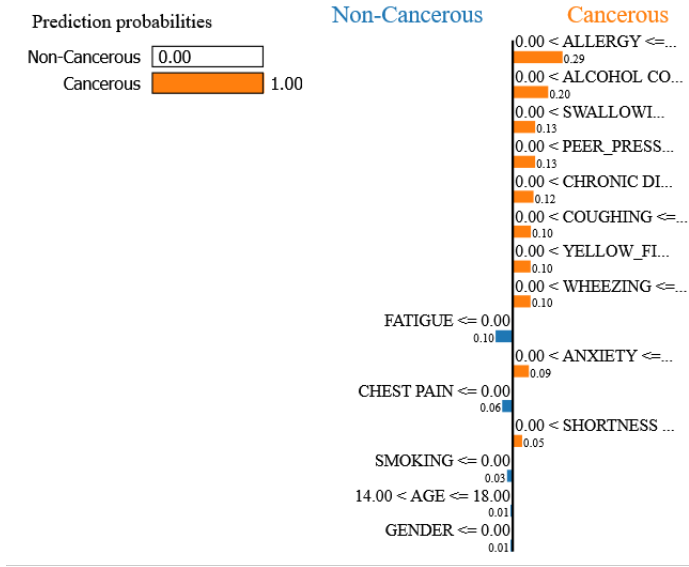


Fig. 5. The Random Forest model's LIME explainability for a single instance (Cancerous)

particular instance as being cancerous or class 1 with 100% confidence. All of these features go towards class 1, which is the cancerous class. Allergy, Alcohol Consuming, Difficulty Swallowing, and Chronic Disease had the biggest effects on the prediction of the cancerous class out of all of these features.

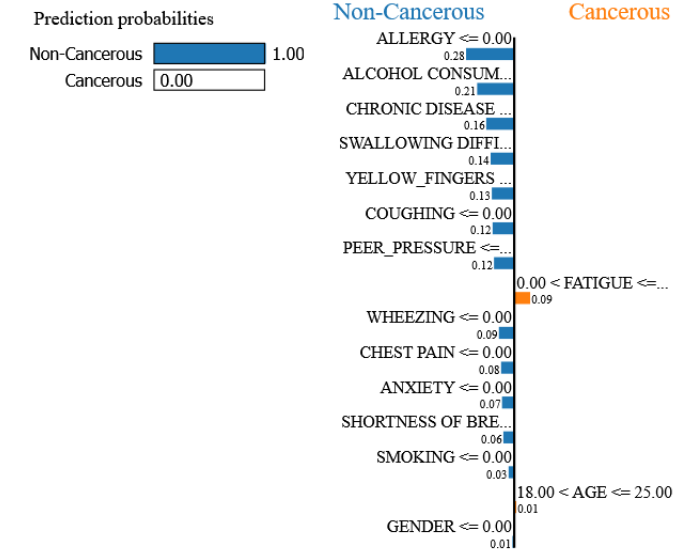


Fig. 6. The Random Forest model's LIME explainability for a single instance (Non-Cancerous)

As seen in Figure 6, the Random Forest model predicts with 100% confidence that this specific instance would be non-cancerous or class 0, and it provides an explanation for the prediction based on all the features aside from fatigue and age. Out of all of these factors, Allergy, Alcohol Consuming, Chronic Disease, and Difficulty Swallowing had the most

influence on the prediction of the non-cancerous class.

According to the preceding explanation, in the instance of SHAP, three features, notably Swallowing Difficulty, Alcohol Consuming, and Coughing, were shared by both ML models when displaying the global interpretation. In the instance of LIME, some key features shared by both ML models included Chronic Disease, Allergy, Swallowing Difficulty, and Alcohol Consuming. As a result, it may be stated that these two sets of features are the most significant for lung cancer detection and interpretation.

V. CONCLUSION

An interpretable lung cancer detection system was successfully developed using four distinct ML models and two XAI models. Among the ML models, Logistic Regression and Random Forest are more accurate in predicting lung cancer, and when combined with SHAP and LIME, they can give interpretability, which improves classification acceptance. Furthermore, a collection of key features for lung cancer diagnosis was discovered. This system can be used to identify lung cancer at an earlier stage more reliably. Deep learning models, along with other XAI models, might be applied to diverse sets of lung cancer datasets in the future. This system could be used to diagnose different illnesses more precisely where model interpretation is a must.

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